How Many Benchmarks Must You Run? Choosing a Few and Inferring the Rest from Leaderboard Score Matrices

Anonymous Author(s)
Affiliation
Address
email

Abstract

A new model is usually evaluated on many benchmarks, each costing compute and time, yet public leaderboards show that benchmark scores are strongly related. We ask: with a budget of k benchmarks out of N, which should be run, how well can the other scores be inferred from a history of models with missing cells, and how large must k be? We propose a Gaussian-conditional predictor fitted to pairwise-complete covariances and a greedy variance-reduction rule that chooses the benchmarks, and give theory: a closed-form error with an irreducible per-benchmark floor, a sandwich $k = \Theta(r/\epsilon)$ for random selection under a rank-r factor model, a spectral lower bound for any selection, and the $k/(M-k-1)$ price of unstructured regression; simulations confirm each. On seven public matrices (MTEB subsets, Open LLM Leaderboard, HELM Lite, LiveBench) we hold out whole models and a fixed set of target benchmarks, so that all selectors are scored on identical targets, with bootstrap CIs. On MTEB (189 models $\times$ 280 tasks), one well-chosen task gives $R^2 = 0.52$ against the mean baseline (random: 0.39), five give 0.71 and ten 0.76; accuracy saturates near 0.80 for every selector, leaving $\sim$7 points of RMSE on single tasks but $\sim$1 point on the average of the remaining scores. Selection reduces MSE by 21% at $k = 1$ and 13% at $k = 10$ versus random and not detectably at $k = 40$; running the most commonly evaluated benchmarks is worse than random. The benefit survives new organisations and later releases at small k, shrinks with sparse or small histories, and vanishes in homogeneous pools and on the smallest LLM matrices. Scoring selectors on the benchmarks left over after selection inflates their advantage.

1 Introduction

Evaluating a model means running it on many benchmarks, and every benchmark costs compute, time and people. Public leaderboards have meanwhile accumulated large model$\times$benchmark score matrices, and the scores of different benchmarks are far from independent. This suggests a concrete budgeted problem. Given a history of M models scored on N benchmarks (with missing cells), a new model can only be run on $k \ll N$ benchmarks: *which k, and how accurately can the remaining $N - k$ scores be inferred? How large must k be?*

Existing efficiency work mostly reduces the number of *items inside a benchmark* [Vivek et al., 2024, Polo et al., 2024b, Kipnis et al., 2024, Perlitz et al., 2024a], and work on relations between benchmarks mostly asks *whether* they agree or lie on a low-dimensional manifold [Perlitz et al., 2024b, Ruan et al., 2024, Polo et al., 2024a]. We study the missing middle: the *choice of whole benchmarks* to run for a new model, with evaluation designed so that the answer is unbiased and comes with uncertainty. We make four contributions.

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1. **A method** (§3): a Gaussian conditional predictor fitted to pairwise-complete covariances of the history, with a greedy variance-reduction rule for choosing S that costs $O(N^2k)$, together with six alternative predictors (mean, averaging, factor model, SVD imputation, nearest neighbours, a blend) and seven alternative selection rules (including random).
2. **Theory** (§3.1, Appendix A): a closed-form error with an irreducible per-benchmark floor (adaptive selection gains nothing under the Gaussian model); a sandwich $\sum_i g_i/(1+kg_i) \le \text{excess} \le r/(k-r-1)$ for random selection, hence $k = \Theta(r/\epsilon)$; a spectral lower bound valid for *any* selection; and the $k/(M-k-1)$ price of unstructured regression on a finite history. All are verified numerically.
3. **Evidence on seven real matrices** (three nested subsets of MTEB, Open LLM Leaderboard, two HELM Lite versions, LiveBench), under random-model, new-organisation and temporal splits, with bootstrap CIs, and ablations of history size, missingness, benchmark pool and score transform.
4. **A protocol point**: scoring each selector on the benchmarks *left over* after its own selection inflates the apparent advantage of informative selection; we score all methods on identical held-out targets.

Findings in brief. On MTEB-main, one well-chosen task gives $R^2 = 0.52$ (random: 0.39), five give 0.71, ten 0.76, and accuracy then saturates near 0.80 for *all* selectors. Selection matters mostly for $k \le 10$ (MSE -21% at $k = 1$, nil at $k = 40$), the most commonly evaluated benchmarks are *worse* than random ones, the benefit shrinks with sparse or small histories, in homogeneous pools and on the smallest LLM matrices, and the fitted Gaussian predicts its own held-out error to within $\sim$10%, which lets the budget be planned from the history.

2 Related work

Cheaper evaluation within one benchmark. Most work on evaluation cost shrinks a *single* benchmark: choosing few items that preserve model rankings or scores, via clustering of model confidences [Vivek et al., 2024], item-response models [Polo et al., 2024b], or a sparse, psychometrically selected item set built from thousands of models' results on six benchmarks [Kipnis et al., 2024]; Perlitz et al. [2024a] frame the general problem of "efficient benchmarking" and study how far a benchmark can be reduced. Our question is orthogonal: which *whole benchmarks* to run, and what they say about the others.

Relations between benchmarks. Large suites report many tasks per model [Hendrycks et al., 2020, Muennighoff et al., 2022, Liang et al., 2022, Zhuo et al., 2024], and judge-based or arena-style evaluations add further axes [Zheng et al., 2023]. Whether benchmarks agree is itself sensitive to methodological choices [Perlitz et al., 2024b]. Observational scaling laws [Ruan et al., 2024] and Sloth [Polo et al., 2024a] exploit the low-dimensional structure of public benchmark scores to predict performance on some benchmarks from others, often as a function of model scale or family. We use no scale or family information, treat the choice of the observed benchmarks as the object of study, and ask how accurate the inference is as a function of the number of benchmarks and of *which* ones.

Subset selection and cold-start elicitation. Choosing k variables to predict the rest is sensor placement in a Gaussian process [Krause et al., 2008], linear subset selection [Das and Kempe, 2011], and column-subset selection. Eliciting a few ratings from a new user is the cold-start problem of recommender systems, studied extensively with active learning [Elahi et al., 2016], and computerized adaptive testing does the same for test-takers. Leaderboards differ in ways that matter: the matrix is small and dominated by one general factor, cells are missing in a model-dependent way, and the new row may be systematically better than the history. We port the Gaussian machinery, state what it implies for k, and test it under these conditions. The predictor side is matrix completion [Mazumder et al., 2010] and probabilistic PCA/factor analysis [Tipping and Bishop, 1999]; covariance estimation uses shrinkage [Ledoit and Wolf, 2004].

3 Problem setting and method

Setting. A history matrix $X \in (\mathbb{R} \cup \{\text{NA}\})^{M \times N}$ holds the scores of M models on N benchmarks, with missing cells. A new model arrives. The evaluator may run any k benchmarks $S \subset [N]$ (the

budget) and must then predict the model's scores on benchmarks it will not run. We standardize every benchmark with its training mean and standard deviation, so that a model's score vector is $z \in \mathbb{R}^N$, and measure error by the *standardized MSE* on the targets (also reported in score points). Two decisions are to be made: *which* S to run (selection), and *how* to infer z_T from z_S (prediction). Both may use only X.

Predictors. Our main predictor treats $z \sim \mathcal{N}(\mu, \Sigma)$ and predicts with the Gaussian conditional mean $\hat{z}_T = \mu_T + \Sigma_{TS}\Sigma_{SS}^{-1}(z_S - \mu_S)$, which is Bayes-optimal under squared loss if the model holds. We estimate Σ from *pairwise-complete* observations of X (so missing cells are handled without imputation), project it to the PSD cone, and shrink it toward the identity on the correlation scale, $\Sigma_\kappa = (1-\kappa)\hat{C} + \kappa I$, with κ chosen on a validation split of the *training* models (GAUSS). We compare against: the benchmark mean (MEAN); the *average of the benchmarks run*, rescaled per target (AVGSHIFT); a rank-r probabilistic factor model $z = \mu + \Lambda f + \varepsilon$, $\varepsilon_j \sim \mathcal{N}(0, \psi_j)$, fitted by EM with missing data [Tipping and Bishop, 1999] and folded in by the posterior of f (FA); iterative-SVD imputation with ridge fold-in in the spirit of soft-impute [Mazumder et al., 2010] (ISVD); K-nearest models (KNN); and a blend of pairwise and factor-model covariance (HYBRID). All hyper-parameters (rank, shrinkage, K) are tuned by an inner split of the training rows.

Selection rules. For a Gaussian model the expected squared error on benchmark j after observing S is $v_j(S) = \Sigma_{jj} - \Sigma_{jS}\Sigma_{SS}^{-1}\Sigma_{Sj} = \mathrm{Var}(z_j \mid z_S)$ and does not depend on the observed values. We choose S to minimize the pool-level objective $F(S) = \sum_j v_j(S)$ (members of S contribute zero), greedily:

$$j_{t+1} \in \arg\max_{j \notin S_t} \frac{\sum_i \mathrm{Cov}(z_i, z_j \mid z_{S_t})^2}{\mathrm{Var}(z_j \mid z_{S_t})} \tag{1}$$

(GREEDY; a rank-one downdate of the residual covariance per step, $O(N^2 k)$ in total, a few milliseconds for $N \approx 300$). Baselines: RANDOM; MOST-EVALUATED (the benchmarks that most models report—what a practitioner would run by default); MOST-CORRELATED (largest mean absolute correlation with the others, which ignores redundancy); k-MEDOIDS on $1 - |\rho|$; GREEDY-MAXVAR (largest conditional variance, the entropy rule of Krause et al. [2008]); GREEDY-MI (the mutual-information rule of Krause et al. [2008]); and top factor-model SNR. Selection is always restricted to the benchmarks the new model has actually been run on (the *pool*), and uses only training rows.

3.1 Theory: what limits accuracy, and how large k must be

All statements are for the Gaussian model; proofs are in Appendix A, and each is checked numerically in §4.5.

Proposition 1 (Closed form, floor, no gain from adaptivity)**.** *If $\Sigma = \Lambda\Lambda^\top + \Psi$ with $\Lambda \in \mathbb{R}^{N \times r}$, then $v_j(S) = \psi_j + \lambda_j^\top \left(I + \Lambda_S^\top \Psi_S^{-1} \Lambda_S\right)^{-1} \lambda_j$. Hence $v_j(S) \geq \psi_j$ for every S (an irreducible per-benchmark floor), v_j is non-increasing in S, and $v_j(S)$ does not depend on the observed scores: choosing benchmarks sequentially, after seeing results, cannot beat choosing the same number in advance.*

Proposition 2 (Random selection: k must scale with the rank)**.** *Let the loading rows be i.i.d. $\mathcal{N}(0, \sigma^2 G)$, $\Psi = \sigma^2 I$, with eigenvalues $g_1 \geq \cdots \geq g_r$ of G, and let S be a uniformly random set of k benchmarks, $j \notin S$. Then*

$$\sum_{i=1}^{r} \frac{g_i}{1 + k g_i} \;\leq\; \frac{\mathbb{E}\, v_j(S)}{\sigma^2} - 1 \;\leq\; \frac{r}{k - r - 1} \qquad (k \geq r + 2).$$

For equal $g_i = g$, the excess error over the floor is therefore $\leq \epsilon\sigma^2$ only if $k \geq r/\epsilon - 1/g$, and is guaranteed once $k \geq r(1 + 1/\epsilon) + 1$: $k = \Theta(r/\epsilon)$. A factor with small strength g_i is not recovered until $k \gtrsim 1/g_i$.

Proposition 3 (No selection beats the spectrum)**.** *For any Gaussian Σ and* any *S with $|S| = k$, $F(S) \geq \sum_{i>k} \lambda_i(\Sigma)$. Under the factor model with $\Psi = \sigma^2 I$ and $k \geq r$ this is $\sigma^2(N - k)$, i.e. $\geq \sigma^2$ per target.*

Proposition 4 (Finite history: unstructured regression pays $k/(M - k - 1)$)**.** *Under Gaussian design with known means, regressing each target on the k selected benchmarks by least squares on $M > k + 1$ complete training rows has $\mathbb{E}[\text{test MSE}] = v_j(S)\,(1 + k/(M - k - 1))$.*

Table 1: Matrices. λ_1/N: top eigenvalue share of the pairwise correlation matrix; PR: participation ratio; n_{90}: components for 90% of the spectrum; ρ: off-diagonal correlation.

Dataset	Models	Bench.	Density	λ_1/N	PR	n_{90}	mean ρ
MTEB-main	189	280	0.77	0.57	2.9	20	0.56
MTEB-dense	175	164	0.90	0.60	2.6	15	0.58
MTEB-eng	185	91	0.70	0.68	2.1	10	0.65
HELM-Lite-22	91	22	1.00	0.58	2.8	8	0.52
HELM-Lite-10	91	10	1.00	0.63	2.3	4	0.55
LiveBench-7	158	7	0.89	0.73	1.8	4	0.68
OpenLLM-LB	4496	6	1.00	0.74	1.8	3	0.67

Greedy quality. $f(S) = \operatorname{tr}\Sigma - F(S) = \sum_j \operatorname{Var}(z_j) R_j^2(S)$ is a non-negative combination of squared multiple correlations, so by Das and Kempe [2011] its submodularity ratio is at least $\lambda_{\min}$ of the correlation matrix and greedy attains $\geq (1 - e^{-\lambda_{\min}})$ of the optimum. This guarantee is weak when benchmarks are strongly collinear (small $\lambda_{\min}$); we therefore measure the greedy/optimal ratio directly (§4.5).

Cost and applicability. Fitting costs one pairwise covariance ($O(MN^2)$) plus tuning; selection is $O(N^2k)$; prediction is one $k \times k$ solve. The history must contain enough co-observations (we require ≥ 5 models per pair, else correlation 0) and the new model is assumed exchangeable with it; §4.3 tests violations.

4 Experiments

Data. All matrices are public and version-pinned (Appendix B.2). **MTEB** [Muennighoff et al., 2022, Embeddings Benchmark contributors, 2026]: one score per (embedding model, task), the task's main metric on its test split (averaged over language subsets; a task enters only with the modal subset signature). From 531 models $\times$ 1,172 tasks (11% filled) we keep tasks with ≥ 100 models and models with ≥ 60 of those tasks: **MTEB-main** (189 models $\times$ 280 tasks, 77% filled, on average 215 observed tasks per model); **MTEB-dense** (≥ 150 models/task, ≥ 80 tasks/model: 175 $\times$ 164, 90%) and **MTEB-eng** (English tasks only: 185 $\times$ 91). LLM leaderboards are small in N: **OpenLLM-LB** (6 benchmarks $\times$ 4,496 models, complete; Hugging Face, 2025), **HELM-Lite-22** (22 scenario/sub-scenario scores $\times$ 91 models; Stanford CRFM, 2025, Liang et al., 2022), its 10-scenario version, and **LiveBench-7** (7 task scores $\times$ 158 models with ≥ 5 of them; White et al., 2024). In every matrix (Table 1) one factor carries 57–74% of the correlation spectrum, but 90% of it needs $n_{90} = 3$–20 components: a dominant general factor plus a long tail.

Protocol. *Rows are held out*, never cells: models are split into 5 folds (3 repeats with different fold seeds for MTEB, 10 for HELM/LiveBench, 1 for OpenLLM), all fitting, tuning and selection use training rows only, and each held-out model is a "new model" that reveals k scores. For every held-out model with observed set C we draw a set T_0 of *held-out targets* (30% of C; two draws per model; for $N \leq 12$ every single benchmark in turn) and let each selector choose S from the pool $C \setminus T_0$. *All selectors and predictors are scored on the same* T_0. Selection is thus restricted to what the model actually has scores for, and neither S nor the baseline determines the targets (§4.6 shows why this matters). Curves compare the same (model, T_0) cells at every k. We report the standardized MSE and $R^2 = 1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{MEAN}}$, where MEAN predicts each target by its training mean; random selection averages 10 draws per cell. Uncertainty: 95% percentile bootstrap over held-out *models*, resampled jointly for paired differences (Wilcoxon p-values are in the released tables). Besides random splits we use *new-organisation* splits (5 folds by organisation prefix, MTEB) and a *temporal* split (training on the oldest 70% of models by release/submission date, testing on the rest: 53 MTEB models, 1,332 OpenLLM models).

4.1 How accurate is inference from k benchmarks, and does selection matter?

Figure 1 and Table 2 give the central result. On MTEB-main the average held-out model has 215 observed tasks, 30% of which (about 65) are the held-out targets, and the mean baseline has RMSE

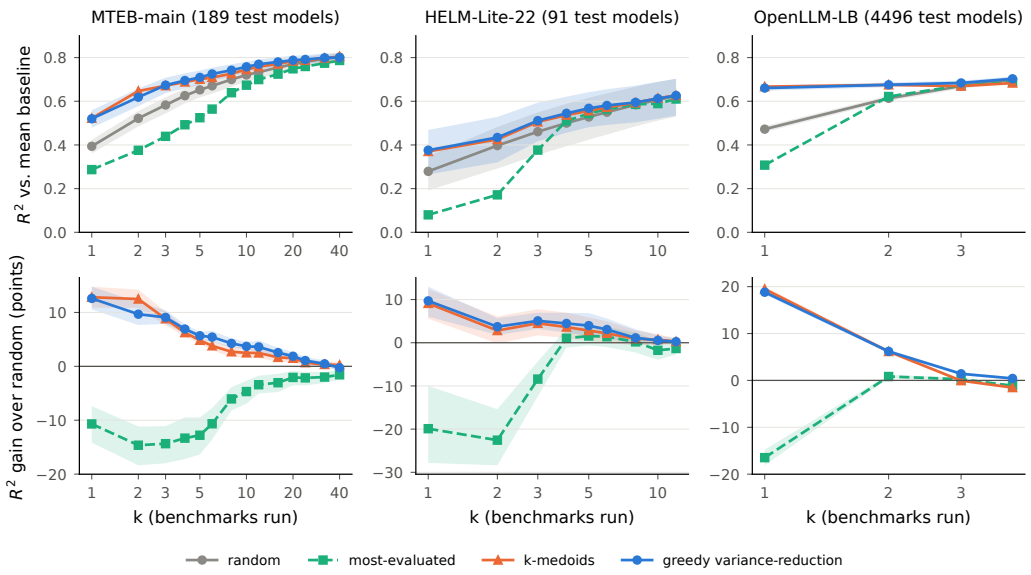


Figure 1: Held-out accuracy vs. number of benchmarks run k (Gaussian predictor). Top: R^2 vs. the mean baseline; bottom: paired R^2 gain over random selection; bands: 95% bootstrap CIs over models. On complete matrices (OpenLLM-LB, HELM) "most-evaluated" is column order. Other matrices: Fig. 4.

Table 2: R^2 [95% CI] at k benchmarks run, Gaussian predictor, random splits of models. Bold: best selector per column (ties at two decimals); "–": k not available for this matrix.

Dataset (split)	Selector	k=1	k=3	k=5	k=10
MTEB-main (rand)	random	0.39 [0.36,0.42]	0.58 [0.54,0.62]	0.65 [0.62,0.68]	0.72 [0.69,0.75]
	most-evaluated	0.29 [0.24,0.33]	0.44 [0.38,0.48]	0.52 [0.47,0.57]	0.67 [0.63,0.71]
	k-medoids	**0.52** [0.48,0.56]	**0.67** [0.64,0.71]	0.70 [0.67,0.73]	0.75 [0.72,0.77]
	greedy variance-reduction	**0.52** [0.48,0.56]	**0.67** [0.64,0.71]	**0.71** [0.68,0.74]	**0.76** [0.73,0.78]
HELM-Lite-22 (rand)	random	0.28 [0.19,0.35]	0.46 [0.35,0.55]	0.53 [0.42,0.62]	**0.61** [0.51,0.69]
	most-evaluated	0.08 [0.04,0.12]	0.38 [0.26,0.46]	0.54 [0.44,0.63]	0.59 [0.49,0.68]
	k-medoids	0.37 [0.26,0.46]	**0.51** [0.42,0.58]	0.56 [0.47,0.63]	**0.61** [0.52,0.69]
	greedy variance-reduction	**0.38** [0.26,0.47]	**0.51** [0.42,0.59]	**0.57** [0.48,0.64]	**0.61** [0.52,0.69]
OpenLLM-LB (rand)	random	0.47 [0.46,0.49]	0.67 [0.66,0.68]	–	–
	most-evaluated	0.31 [0.29,0.33]	0.67 [0.66,0.68]	–	–
	k-medoids	**0.67** [0.66,0.68]	0.67 [0.66,0.68]	–	–
	greedy variance-reduction	0.66 [0.65,0.67]	**0.68** [0.67,0.69]	–	–
LiveBench-7 (rand)	random	0.47 [0.40,0.53]	0.61 [0.54,0.67]	–	–
	most-evaluated	0.50 [0.41,0.58]	**0.64** [0.58,0.69]	–	–
	k-medoids	**0.53** [0.46,0.60]	0.60 [0.54,0.66]	–	–
	greedy variance-reduction	**0.53** [0.45,0.59]	0.59 [0.53,0.65]	–	–

17.4 score points per target. With *one* well-chosen task, $R^2 = 0.52$ (RMSE 11.0 points); with five, 0.71 (8.0 points); with ten, 0.76 (7.2 points). After that returns diminish quickly: at $k = 40$ both greedy and random reach $R^2 \approx 0.80$ (6.7 vs. 6.4 points). The plateau is reached by every selector and by the best predictors (Appendix Table 5): the unexplained 20% is consistent with benchmark-specific variation that the other benchmarks do not carry (we cannot separate it from evaluation noise). It is far smaller for *aggregates*: the mean of the held-out scores is predicted with mean absolute error 1.7 points at $k = 5$ and 1.3 at $k = 10$ (random: 2.7 and 1.8; mean baseline: 10.5), because benchmark-specific errors average out (Appendix B.3).

Selection matters most at the smallest budgets. On MTEB-main, greedy selection lowers the MSE of random selection by 21% [17, 25] at $k = 1$, 16% [14, 19] at $k = 5$, 13% [12, 15] at $k = 10$, 8% [6, 11] at $k = 20$, and not detectably at $k = 40$ (-1% [-4, 1]); the number of tasks needed for $R^2 \geq 0.75$ falls from 16 to 10 (Table 3). k-medoids on $1 - |\rho|$ is nearly as good (equal or better

Table 3: Smallest k reaching a target R^2 ("$> K$": not reached up to the largest k evaluated).

Dataset (protocol)	Selector	$R^2 \geq 0.6$	$R^2 \geq 0.7$	$R^2 \geq 0.75$
HELM-Lite-22 (rand)	random	10	>12	>12
	greedy variance-reduction	10	>12	>12
MTEB-dense (rand)	random	4	10	20
	greedy variance-reduction	2	6	16
MTEB-eng (rand)	random	2	5	10
	greedy variance-reduction	1	3	5
MTEB-main (org)	random	6	20	>40
	greedy variance-reduction	3	10	>40
MTEB-main (rand)	random	4	10	16
	greedy variance-reduction	2	5	10
MTEB-main (time)	random	12	>40	>40
	greedy variance-reduction	6	>40	>40
OpenLLM-LB (rand)	random	2	>4	>4
	greedy variance-reduction	1	4	>4
OpenLLM-LB (time)	random	2	>4	>4
	greedy variance-reduction	1	4	>4

at $k \leq 3$, 3–5% worse for $k \in \{5, 10\}$); the max-variance ("entropy") rule of Krause et al. [2008] is much worse at small k (it picks idiosyncratic tasks) and their mutual-information rule is slightly worse than ours; choosing the benchmarks with the largest average correlation is as good as greedy at $k = 1$ but *worse than random* from $k = 6$ on (-9.9% [-13.8, -6.3] at $k = 10$), because it picks redundant, mutually correlated benchmarks. Most strikingly, running *the most commonly reported tasks* is worse than random at every k we tried: -18% [-24, -12] at $k = 1$ and -37% [-48, -27] at $k = 5$ (MTEB-main, where popularity is informative). Across matrices (Table 2, Fig. 4) the MSE reduction at $k = 1$ is 12–36% and shrinks with k. In the three tiny LLM matrices the useful range of k is short: OpenLLM-LB: one well-chosen benchmark gives $R^2 = 0.66$ vs. 0.47 (random), three give 0.68 vs. 0.67, and the plateau is 0.70; HELM-Lite-22 greedy gain is 13% [8, 19] at $k = 1$ and not significant at $k = 10$ (1% [-1, 4]); on LiveBench-7 selection is not helpful beyond $k = 1$ and greedy is *worse* than random at $k = 3$ (-4% [-7, -2]).

4.2 Which predictor?

Appendix Table 5 fixes the selection (random or greedy) and varies the predictor (hyper-parameters tuned on the training rows). The naive "average of what was run" (AVGSHIFT) saturates at $R^2 \approx 0.6$: no amount of averaging substitutes for modelling which benchmarks resemble which. The covariance-based predictors are close to each other, and no predictor is uniformly best (paired CIs in Appendix Table 8). HYBRID is slightly but significantly better than GAUSS for $k \geq 10$ (+0.004 to +0.023 R^2, largest at $k = 40$), FA and ISVD are 0.01–0.04 behind it for $k \geq 10$ (the tuned factor rank is 3–12, so the covariance is *not* well described by one or two factors), and KNN is *better* at $k = 1$ (+0.05 [0.03, 0.07] under random and +0.08 [0.05, 0.10] under greedy selection) and about equal for $k = 5$–20; a single benchmark apparently has a non-linear relation to the others that a linear-Gaussian conditional misses. Greedy selection, which assumes the Gaussian model, improves every covariance-based predictor and KNN too.

4.3 Does it survive distribution shift?

Table 4 repeats the MTEB-main experiment with *new-organisation* and *later-release* test models. The mean baseline degrades (its standardized MSE is 1.36 for new organisations and 3.25 for later releases vs. 1.18: later models differ systematically from the history, presumably by scoring higher), and R^2 at $k = 40$ drops to ≈ 0.72 and ≈ 0.65. The benefit of selection survives at small budgets (new organisation: 21%, 17%, 22%, 18% MSE reduction at $k = 1, 2, 3, 5$, all CIs exclude 0), vanishes at $k = 20$ for new organisations (1% [-3, 5]), and persists for later releases (13% [8, 17] at $k = 20$). Selection from the history alone remains valuable at $k \leq 5$ even when the new model comes from an unseen lab; but the budget needed for a given R^2 increases ($R^2 \geq 0.7$ needs 10 tasks for new organisations, and > 40 for later releases, Table 3).

Table 4: MTEB-main under three protocols (R^2, Gaussian predictor).

Protocol	Selector	k=1	k=3	k=5	k=10	k=20
random models	random	0.39	0.58	0.65	0.72	0.77
	most-evaluated	0.29	0.44	0.52	0.67	0.75
	k-medoids	0.52	0.67	0.70	0.75	0.78
	greedy variance-reduction	0.52	0.67	0.71	0.76	0.79
new organisation	random	0.35	0.54	0.60	0.66	0.71
	most-evaluated	0.28	0.32	0.45	0.61	0.65
	k-medoids	0.50	0.62	0.66	0.68	0.71
	greedy variance-reduction	0.49	0.64	0.67	0.70	0.71
later release	random	0.31	0.51	0.55	0.59	0.64
	most-evaluated	0.25	0.11	0.39	0.54	0.61
	k-medoids	0.47	0.55	0.60	0.64	0.66
	greedy variance-reduction	0.46	0.52	0.60	0.65	0.68

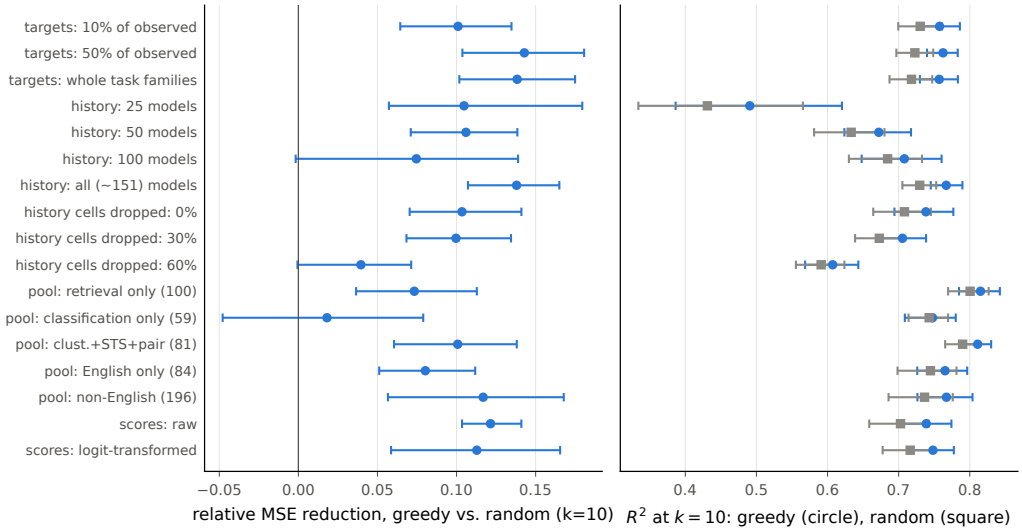


Figure 2: Data conditions (MTEB-main, $k = 10$). Left: relative MSE reduction of greedy vs. random with 95% CI. Right: R^2 of greedy (circle) and random (square).

4.4 Under which data conditions do the conclusions hold?

Figure 2 varies one data property of MTEB-main at a time. (i) *History size*: the greedy gain over random is 8–14% for every history size (10% [6, 18] with 25 training models, 14% [11, 17] with all ∼151; CIs of ≤ 100 models include small values), but absolute accuracy changes a lot: greedy R^2 at $k = 10$ is 0.49, 0.67, 0.71, 0.77 for 25, 50, 100, all training models. (ii) *Missingness*: dropping 30% of the history's observed cells leaves the gain at 10%, dropping 60% cuts it to 4% [0, 7] and R^2 from 0.74 to 0.61 (greedy at $k = 10$). (iii) *Pool composition*: greedy helps in every pool except when benchmarks are drawn only from the 59 classification tasks (gain 2% [−5, 8]); the gain is largest for the non-English pool (12% [6, 17]) and the clustering+STS+pair pool (10% [6, 14]). (iv) A *logit transform* of the scores does not help ($R^2 = 0.75$ vs. 0.74 at $k = 10$, CIs overlap; both evaluated in score points). (v) *Target construction*: holding out 10% or 50% of each model's observed benchmarks, or whole name-based task families (so that no near-duplicate of a target is in the pool), leaves the $k = 10$ gain at 10% [6, 14], 14% [10, 18] and 14% [10, 18]. (vi) Estimating the selection covariance from the training rows costs little: using a covariance estimated with the test models included ("oracle") improves the MSE at $k \leq 10$ by only 1–4% over our greedy rule, whereas selecting with the low-rank factor-model covariance is 5–9% *worse* at $k = 10, 20$ on MTEB-main (Appendix Table 9).

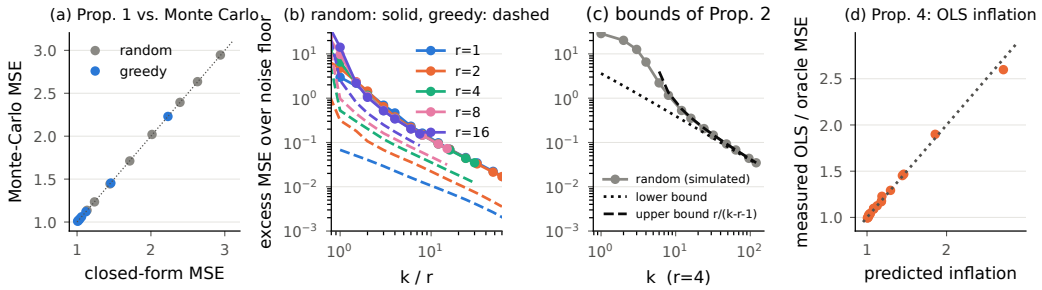


Figure 3: Synthetic checks. (a) Prop. 1 vs. Monte Carlo. (b) Excess error over the noise floor vs. k/r (N=400, $g_i = 10$): random-S curves for different ranks collapse. (c) Random-S excess ($r = 4$) between the bounds of Prop. 2. (d) OLS/oracle MSE ratio vs. $1 + k/(M - k - 1)$ (Prop. 4).

4.5 Theory against experiment

Synthetic factor models (Fig. 3). The closed form of Prop. 1 matches Monte Carlo to within 0.006 in MSE for all random and greedy sets tested. The sandwich of Prop. 2 holds in all 94 (rank, strength, k) cells with $k \geq r + 2$ (within 3 MC standard errors), and the smallest k with simulated excess $\leq \epsilon\sigma^2$ lies in $[r/\epsilon - 1/g,\ r(1 + 1/\epsilon) + 1]$ in 21 of 24 settings and exceeds the upper end by one (MC noise at the bound) in the other three: e.g. $r = 8,\ \epsilon = 0.1$ requires $k = 90$, i.e. $\approx 11r$. Greedy selection reaches the same excess with 0.08–0.48 (median 0.29) times as many benchmarks (e.g. 38 instead of 90 for $r = 8, \epsilon = 0.1$), a far larger gain than on real matrices, whose spectra have one dominant factor that random sets already capture. The OLS inflation $1 + k/(M - k - 1)$ of Prop. 4 agrees with simulation to a mean absolute deviation of 0.016 (max. 4.3% relative). The spectral bound of Prop. 3 is never violated by the exhaustive optimum (0 of 480 instances) and is on average 82% of it. Greedy selection recovers 99.7% of the variance reduction of the exhaustive optimum on average (min. 0.91, 480 small instances; random: 0.87; the Das–Kempe bound is 0.18 on average, hence loose), and a swap local search from the greedy set improves the residual objective by 1.4% on average (max. 13%) at $N = 60$. The gain of greedy over random depends on the structure as the theory suggests (Appendix Table 10): at $k = 8$ the reduction of the residual objective is 3–8% for one factor, 4–16% for two, 5–29% for four and 10–57% for eight; it grows with factor strength g and rank r and shrinks when leverage is heterogeneous (log-sd κ of the loading scales).

Real matrices. The error that the fitted Gaussian predicts for itself, $\mathrm{tr}(\Sigma_{T|S})/|T|$, tracks the held-out MSE closely for both selectors (Appendix Fig. 5): its ratio to the held-out error averages 0.97 (MTEB-main), 0.89 (MTEB-dense), 1.10 (MTEB-eng), 1.07 (HELM-22), and 1.03 (OpenLLM), within 0.62–1.26 over all k and matrices. The model-implied curve can therefore be used to *plan* k for a target accuracy before running anything. Prop. 3 bounds in-sample residuals of any selection from below (Appendix Fig. 6); the in-sample greedy residual is 1.2–1.6× the bound on MTEB-main for $k \leq 40$, while random selection is 1.5–2.0×.

4.6 A protocol pitfall: scoring the complement

A natural protocol lets each selector pick S from *all* observed benchmarks and scores the model on the *remaining* $C \setminus S$. Then selectors are scored on different targets, and a rule that removes the hardest-to-explain benchmarks into S leaves the easy ones as targets. Appendix Table 6 compares both protocols for the same selected sets. With complement targets greedy appears to beat random at $k = 40$ on MTEB-main by a MSE of 0.18 vs. 0.23 and at $k = 12$ on HELM-Lite-22 by 0.27 vs. 0.42; on fixed targets the advantage is nil (0.24 vs. 0.24 and 0.41 vs. 0.42). The inflation is largest exactly where the budget is large relative to N.

5 Discussion and limitations

What the evidence supports. (1) A handful of well-chosen benchmarks gives most of what a larger budget would: on MTEB-main five tasks of ∼215 explain 71% of the variance (relative to the mean baseline) of a new model's other scores, ten explain 76%, and no budget up to 40 passes ≈80%.

(2) The choice of benchmarks matters mainly below $k \approx 10$ and the gain over random is substantial there (13–21% in MSE), but it is not uniformly positive: it is absent for homogeneous pools and for k comparable to the pool size, and it is negative for LiveBench-7 at $k = 3$. (3) Heuristics that look reasonable can hurt: the most-evaluated benchmarks are worse than random at every k we tried on MTEB-main, and the most-correlated ones from $k = 6$ on. (4) The theory explains the shape: an irreducible per-benchmark floor, a requirement $k \gtrsim r(1 + 1/\epsilon)$ for a factor structure of rank r, and a spectral lower bound that no selection can beat; the sandwich and the finite-history inflation are accurate in simulation, and the fitted Gaussian predicts its own error on real matrices. The hypothesis that a small subset suffices to infer the rest is thus *supported in the regime of moderate accuracy* (about 75% of variance) and *not supported for accurate prediction of individual scores*, where a plateau remains.

Where it fails or is unproven. **Plateau.** A residual of 20–30% of variance is benchmark-specific; we do not know how much is evaluation noise and how much true specialization, so we cannot say whether more data or better predictors would remove it. **Gaussian linear model.** The mean-and-covariance model ignores ceiling effects and non-linear dependence; kNN beats it at $k = 1$ by 5–8 points of R^2, and a logit transform gave no gain. The factor-model covariance, although natural for the theory, selected 5–9% worse than the pairwise covariance at $k \geq 10$, and low-rank prediction was never better than the full covariance: our theory is a guide to scaling, not a description of these matrices. **Selection is not adaptive.** Under the Gaussian model adaptivity gives nothing (Prop. 1), but we did not test adaptive policies with non-Gaussian predictors. **Retrospective protocol.** A model's observed benchmarks are not random: authors choose where to report (missing not at random); we restrict selection to the model's observed set and score on its other observed benchmarks, which may flatter methods in either direction. Costs per benchmark are ignored; a cost-aware greedy ratio rule is a one-line change but untested. **Data.** MTEB models are text-embedding models, not generative LLMs; the three LLM matrices have $N \leq 22$, so k cannot exceed 3–12 and conclusions about large budgets rest on MTEB alone. Near-duplicate models (merges, fine-tunes, sizes of one family) make random splits optimistic; the organisation and temporal splits reduce but do not remove this. The time split for MTEB has only 53 test models. The MTEB subset we use consists of tasks evaluated for $\geq$100 models, which over-represents widely run tasks. We tuned hyper-parameters on the training rows only, but chose GAUSS as the default after looking at inner validation on MTEB-main, and the study designed its protocol after a first, flawed run with complement targets (§4.6; that section reports a controlled re-run, not the first run). **Scope of the claims.** Statements about "k needed" depend on the target accuracy, the matrix, and the shift between history and new model; they are not universal constants.

Practical guidance. With a history like MTEB-main, run 5–10 benchmarks chosen by greedy variance reduction (or k-medoids), predict with the conditional Gaussian (kNN for $k = 1$), expect the *average* of the other scores to be within $\sim$1–2 points but individual scores only within $\sim$7–8, plan k with the model-implied error, and re-check against full evaluations when the new model comes from a different lab or period.

Broader impact. Cheaper evaluation saves energy and lowers entry barriers, but risks over-reliance on proxy benchmarks; inferred scores should be labelled as such.

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A Proofs of the propositions

Throughout, $z \sim \mathcal{N}(0, \Sigma)$ is standardized; S is fixed before the data are seen (the posterior covariance below does not depend on the data, which is the content of the last claim of Prop. 1).

Proof of Prop. 1. Write $z = \Lambda f + \varepsilon$, $f \sim \mathcal{N}(0, I_r)$, $\varepsilon \sim \mathcal{N}(0, \Psi)$. Given z_S, $f \mid z_S \sim \mathcal{N}(m, P)$ with $P = (I + \Lambda_S^\top \Psi_S^{-1} \Lambda_S)^{-1}$ and $m = P\Lambda_S^\top \Psi_S^{-1} z_S$ (Gaussian conditioning). For $j \notin S$, $z_j = \lambda_j^\top f + \varepsilon_j$ with ε_j independent of (f, z_S), so the Bayes predictor is $\lambda_j^\top m$ and $\mathrm{Var}(z_j \mid z_S) = \psi_j + \lambda_j^\top P \lambda_j$. This equals $\Sigma_{jj} - \Sigma_{jS}\Sigma_{SS}^{-1}\Sigma_{Sj}$ by the Woodbury identity. Since $P \succeq 0$, $v_j \geq \psi_j$. Adding an index s to S adds $\lambda_s \lambda_s^\top / \psi_s \succeq 0$ to P^{-1}, hence lowers P in the Loewner order and v_j weakly decreases. P is free of z_S, so for any (possibly adaptive) selection policy the conditional variance after k steps is that of the set actually chosen; the expected error of an adaptive policy is an average of $v_j(S)$ over the sets it can produce, which is at least $\min_{|S|=k} v_j(S)$, attained by a non-adaptive set. □

Proof of Prop. 2. Let $W = \Lambda_S^\top \Lambda_S / \sigma^2 = \sum_{s \in S} \tilde{\lambda}_s \tilde{\lambda}_s^\top$ with $\tilde{\lambda}_s \sim \mathcal{N}(0, G)$ i.i.d., i.e. $W \sim \mathrm{Wishart}_r(k, G)$, so $\mathbb{E}W = kG$, and for $k \geq r + 2$, $\mathbb{E}W^{-1} = G^{-1}/(k - r - 1)$. For the target $j \notin S$, λ_j is independent of S with $\mathbb{E}\lambda_j \lambda_j^\top = \sigma^2 G$, hence $\mathbb{E}[v_j - \sigma^2] = \sigma^2 \operatorname{tr}\big(G\,\mathbb{E}(I + W)^{-1}\big)$. *Lower bound:* $W \mapsto (I+W)^{-1}$ is operator convex, so $\mathbb{E}(I+W)^{-1} \succeq (I+kG)^{-1}$ and $\operatorname{tr}(G(I+kG)^{-1}) = \sum_i g_i/(1 + kg_i)$. *Upper bound:* $(I + W)^{-1} \preceq W^{-1}$ whenever $W \succ 0$ (a.s. for $k \geq r$), so $\operatorname{tr}(G\,\mathbb{E}W^{-1}) = r/(k - r - 1)$. For $g_i \equiv g$: the lower bound is $rg/(1 + kg) \leq \epsilon$ iff $k \geq r/\epsilon - 1/g$, and the upper bound is $\leq \epsilon$ iff $k \geq r + 1 + r/\epsilon$. □

Proof of Prop. 3. $\Sigma_{\cdot|S} = \Sigma - B$ with $B = \Sigma_{\cdot S}\Sigma_{SS}^{-1}\Sigma_{S\cdot} \succeq 0$ of rank $\leq k$ and $F(S) = \operatorname{tr}(\Sigma - B)$. By Weyl's inequality, $\lambda_{i+k}(\Sigma) \leq \lambda_i(\Sigma - B) + \lambda_{k+1}(B) = \lambda_i(\Sigma - B)$, so $\operatorname{tr}(\Sigma - B) \geq \sum_{i \geq 1} \lambda_{i+k}(\Sigma)$. For $\Sigma = \Lambda\Lambda^\top + \sigma^2 I$ with $\operatorname{rank}\Lambda = r$, $\lambda_i(\Sigma) = \sigma^2$ for $i > r$, so for $k \geq r$ the sum is $(N - k)\sigma^2$. □

Proof of Prop. 4. Under Gaussianity, $z_j = \beta^\top z_S + e$ with $e \sim \mathcal{N}(0, s^2)$ independent of z_S and $s^2 = v_j(S)$. With $X_S \in \mathbb{R}^{M \times k}$ the training design, $\hat{\beta} - \beta \mid X_S \sim \mathcal{N}(0, s^2(X_S^\top X_S)^{-1})$. For a fresh z_S, $\mathbb{E}(\hat{\beta}^\top z_S - z_j)^2 = s^2 + \mathbb{E}\, z_S^\top (\hat{\beta} - \beta)(\hat{\beta} - \beta)^\top z_S = s^2\big(1 + \operatorname{tr}(\Sigma_{SS}\,\mathbb{E}(X_S^\top X_S)^{-1})\big)$. As $X_S^\top X_S \sim \mathrm{Wishart}_k(M, \Sigma_{SS})$, $\mathbb{E}(X_S^\top X_S)^{-1} = \Sigma_{SS}^{-1}/(M - k - 1)$, giving $s^2(1 + k/(M - k - 1))$. □

On the greedy guarantee. $f(S) = \sum_j \mathrm{Var}(z_j) R_j^2(S)$ with $R_j^2(S) = 1 - v_j(S)/\Sigma_{jj}$ (equal to 1 if $j \in S$). Das and Kempe [2011] show that the objective $R^2(y; z_S)$ of selecting S to linearly predict a single variable y has submodularity ratio $\gamma \geq \lambda_{\min}(C)$, C the correlation matrix of the candidates, and that greedy forward selection attains at least $(1 - e^{-\gamma})$ of the optimum. The submodularity ratio of a non-negative combination of monotone set functions is at least the minimum of the ratios, so the same bound holds for f (we rely on their theorem for the single-target case and do not re-derive it). Under the factor model $\lambda_{\min}(C) \geq \min_j \psi_j/\Sigma_{jj}$ (since $\Sigma \succeq \Psi$), which is small whenever some benchmark is nearly explained by the others; the guarantee is then loose, and the empirical check of §4.5 is the operative evidence.

B Further experimental details and tables

B.1 Implementation details

Standardization. Per benchmark, with training-row mean and standard deviation (observed cells only). Errors are always reported on the standardized scale ("MSE") and, where stated, in score points; $R^2 = 1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{MEAN}}$ with MEAN the training-mean predictor on the same targets (its standardized MSE exceeds 1 because held-out models differ systematically from the history: 1.18 on random splits, 1.36 for new organisations, 3.25 for later releases, MTEB-main). **Covariance.** Pairwise-complete correlation with ≥ 5 co-observed models per pair (else 0), eigenvalues clipped at 10^{-3} and rescaled to unit diagonal, then $\Sigma_\kappa = (1 - \kappa)\hat{C} + \kappa I$ with $\kappa \in \{0, 0.05, 0.15, 0.3, 0.5\}$ chosen on an inner split (20% of training rows as validation models; for each, a random half of its observed benchmarks (at most 10) plays S, the rest targets). The same inner split tunes FA rank $r \in \{1, 2, 3, 4, 6, 8, 12\}$ (restricted to $(N - r)^2 > N + r$), ISVD rank/ridge, K of kNN ($\{5, 15, 40\}$) and the HYBRID blend; GAUSS was selected for the main pipeline *because* it did best in that inner validation on MTEB-main. **FA by EM** uses the missing-data E-step (per-row posterior over observed benchmarks), a small ridge on loadings (0.05), and a floor $\psi_j \geq 0.02$ on unique variances; 300

iterations at most. **Selectors.** GREEDY is Eq. (1); GREEDY-MI maximizes $\mathrm{Var}(z_j \mid z_S)\,(\Sigma^{-1})_{jj}$ (the criterion of Krause et al. [2008] simplifies since $\mathrm{Var}(z_j \mid \text{all others}) = 1/(\Sigma^{-1})_{jj}$ does not depend on S); k-MEDOIDS: 5 k-means++-seeded restarts on $1 - |\rho_{ij}|$, medoids are not nested in k; "top-SNR" ranks by factor-model communality over unique variance. "Most-evaluated" orders benchmarks by their number of training observations (ties by column order). **Held-out targets.** T_0 is a random 30% of the model's observed set (at least 2) with two draws per model for MTEB, four for HELM-22 (pool sizes: up to ∼195 on MTEB-main); for $N \leq 12$ the held-out target is each benchmark in turn. We evaluate $k \leq 0.7\,|C| - 3$ roughly, capped at 40 (14 for HELM-22; $\min_i |C_i| - 2$ for $N \leq 12$). A (model, target-draw) cell is used in a curve only if it is available for every k of that curve. **Aggregation.** For each (selector, predictor, k), values are averaged over random draws and cells, then over repeats/folds per model; the bootstrap resamples models (2000 replicates, seed fixed).

B.2 Reproducibility

Code, the processed matrices' construction scripts, and a README with commands, seeds, timings and the script that generates every table and figure are in the supplementary archive. **Pinned data:** MTEB results repository at commit `6dabbcc6` (CC0-1.0); Open LLM Leaderboard `contents` at `9c09a7ca` (no explicit license; 4,576 rows, we drop 80 flagged or duplicate model entries); HELM Lite release v1.13.0 (`core_scenarios`, `mmlu`, `legalbench`, `wmt_14` group files); LiveBench `model_judgment` at `9704e5da` (a 7-task per-model summary; mean of per-question scores per model and task $\times 100$). MTEB task types and languages come from the `mteb` package (v2.26.31; used only for task metadata, 958 of the 1,172 tasks matched; the others are typed by name). One MTEB score is the `main_score` of the test split (validation or dev if no test split), averaged over language subsets; tasks with differing subset sets across models keep only the modal one; tasks with scores outside $[-100, 100]$ (two hardware-speed tasks) are dropped. **Compute:** CPU only (40 cores, 250 GB RAM, of which a few GB used). The full pipeline (`run_everything.sh`: synthetic checks, all real-data runs, ablations, tables) took about 1.1 h wall-clock; the second stage ran up to ∼80 processes on the 40 cores, so the summed per-fold wall times (22.7 h) overstate CPU time; one MTEB-main random-split fold (tuning of 6 predictors + all selectors) takes ≈ 8 min on one core when the machine is not oversubscribed, the synthetic checks ≈ 5 min. Data: 3.4 GB for the cloned MTEB repository (extracted matrices: a few MB).

B.3 Additional results

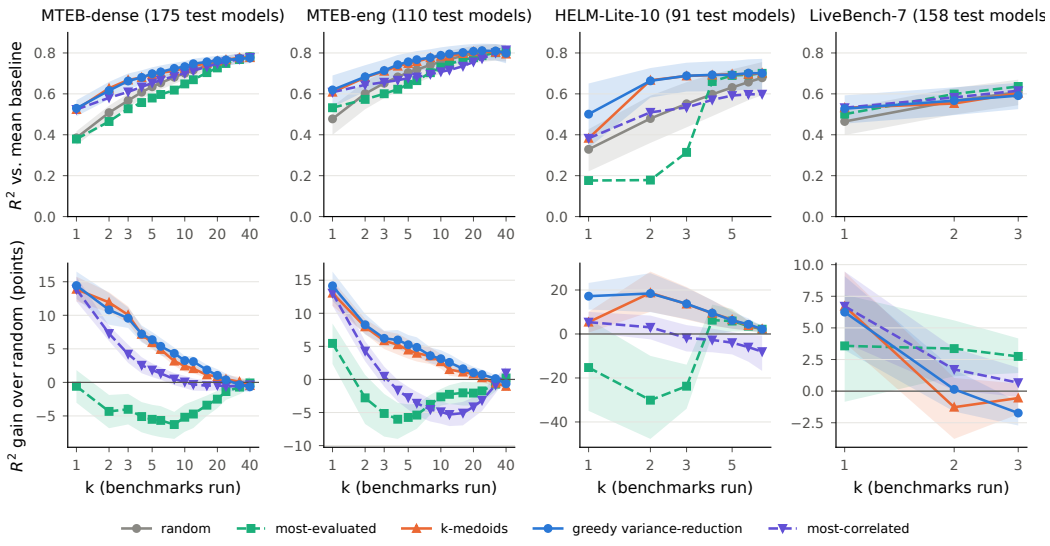


Figure 4: As Fig. 1 for the other matrices (random splits), with the most-correlated selector added. HELM-Lite-10 and LiveBench-7 have complete or nearly complete pools; the number of test models is the number of held-out models that have all budgets available.

Table 5: Predictors on MTEB-main (R^2; random splits). 95% CIs of $\pm$0.03–0.04 for each cell; paired differences in Appendix Table 8.

Selection	Predictor	k=1	k=5	k=10	k=20	k=40
random	AvgShift	0.40	0.55	0.58	0.60	0.61
	kNN	0.45	0.67	0.74	0.77	0.78
	ISVD	0.40	0.64	0.70	0.74	0.77
	FA	0.38	0.65	0.71	0.76	0.78
	Gauss	0.39	0.65	0.72	0.77	0.80
	Hybrid	0.39	0.65	0.72	0.77	0.82
greedy variance-reduction	AvgShift	0.53	0.58	0.59	0.59	0.60
	kNN	0.60	0.70	0.75	0.77	0.78
	ISVD	0.53	0.69	0.74	0.75	0.76
	FA	0.54	0.71	0.75	0.77	0.79
	Gauss	0.52	0.71	0.76	0.79	0.80
	Hybrid	0.52	0.71	0.76	0.80	0.82

Table 6: Standardized MSE under fixed held-out targets T_0 (this paper) vs. complement targets $C \setminus S$ (naive).

Dataset	Targets	Selector	small k	mid k	large k
HELM-Lite-22 (k=2, 5, 12)	fixed T_0	random	0.67	0.54	0.42
HELM-Lite-22 (k=2, 5, 12)	fixed T_0	greedy variance-reduction	0.63	0.48	0.41
HELM-Lite-22 (k=2, 5, 12)	complement $C \setminus S$	random	0.66	0.48	0.42
HELM-Lite-22 (k=2, 5, 12)	complement $C \setminus S$	greedy variance-reduction	0.61	0.42	0.27
MTEB-main (k=2, 10, 40)	fixed T_0	random	0.57	0.34	0.24
MTEB-main (k=2, 10, 40)	fixed T_0	greedy variance-reduction	0.47	0.29	0.24
MTEB-main (k=2, 10, 40)	complement $C \setminus S$	random	0.55	0.33	0.23
MTEB-main (k=2, 10, 40)	complement $C \setminus S$	greedy variance-reduction	0.45	0.27	0.18

Aggregate error. Mean absolute error (score points) of the *average* of the held-out scores of a model (MTEB-main, GAUSS, random splits; 95% bootstrap CI): greedy 3.8 [3.5, 4.2] at $k = 1$, 2.1 [1.9, 2.3] at 3, 1.7 [1.6, 1.9] at 5, 1.3 [1.2, 1.4] at 10, 1.1 [1.0, 1.2] at 20; random 6.2 [5.9, 6.5], 3.7 [3.5, 3.8], 2.7 [2.6, 2.8], 1.8 [1.7, 1.9], 1.3 [1.2, 1.4]; mean baseline 10.5. The RMSE per individual held-out benchmark is 11.0, 8.7, 8.0, 7.2, 6.9 (greedy) and 17.4 (baseline).

Ranking quality. The per-benchmark Spearman correlation between predicted and actual scores across held-out models (MTEB-main, GAUSS; average over benchmark$\times$fold pairs with ≥ 8 models) is 0.79/0.81/0.82 for greedy and 0.76/0.78/0.81 for random selection at $k = 5/10/20$; MTEB-dense 0.77/0.78/0.79 vs. 0.73/0.77/0.79; HELM-Lite-22 0.71/0.72 vs. 0.68/0.74 at $k = 5/10$. Models are therefore ranked well on any single benchmark with a handful of runs, and selection helps ranking less than it helps error.

Stability of the selected sets. Running the greedy rule on each of the 15 training folds of the MTEB-main random splits (all 280 tasks as candidates, default shrinkage 0.05), the selected sets overlap only moderately: mean pairwise Jaccard similarity 0.23, 0.26, 0.26 at $k = 5, 10, 20$ (a random set would give 0.01, 0.02, 0.04), because many tasks are near-substitutes. The most frequent first-ten picks are CyrillicTurkicLangClassification (13 of 15 folds), STS14 (12), TempReasonL2Fact (10), CQADupstackPhysicsRetrieval (10), AngryTweetsClassification (8) and BelebeleRetrieval (8). A practitioner should therefore treat the greedy list as one of many near-equivalent good subsets.

Number of models in the evaluation. After requiring all budgets to be available for a cell, MTEB-main random splits use 189 models and 1,134 (model, fold, target-draw) cells, MTEB-main time-split 53 models; MTEB-eng 110 models; OpenLLM 4,496 (random) and 1,332 (temporal); HELM 91; LiveBench 158 in the first-3 budgets.

Table 7: As Table 2 for the other matrices.

Dataset (split)	Selector	k=1	k=3	k=5	k=10
MTEB-dense (rand)	random	0.38 [0.35,0.42]	0.57 [0.53,0.60]	0.63 [0.60,0.67]	0.70 [0.67,0.73]
	most-evaluated	0.38 [0.34,0.42]	0.53 [0.49,0.57]	0.58 [0.54,0.61]	0.65 [0.62,0.68]
	k-medoids	0.52 [0.48,0.56]	**0.67** [0.63,0.70]	0.69 [0.66,0.72]	**0.73** [0.69,0.75]
	greedy variance-reduction	**0.53** [0.49,0.57]	0.66 [0.62,0.70]	**0.70** [0.66,0.73]	**0.73** [0.70,0.76]
MTEB-eng (rand)	random	0.48 [0.40,0.54]	0.65 [0.58,0.71]	0.70 [0.64,0.76]	0.76 [0.70,0.80]
	most-evaluated	0.53 [0.45,0.60]	0.60 [0.52,0.66]	0.65 [0.58,0.70]	0.73 [0.67,0.78]
	k-medoids	0.61 [0.53,0.67]	**0.71** [0.64,0.76]	0.75 [0.69,0.79]	0.78 [0.73,0.82]
	greedy variance-reduction	**0.62** [0.53,0.69]	**0.71** [0.64,0.77]	**0.76** [0.70,0.80]	**0.79** [0.74,0.83]
HELM-Lite-10 (rand)	random	0.33 [0.22,0.43]	0.55 [0.44,0.66]	0.63 [0.53,0.72]	–
	most-evaluated	0.18 [0.06,0.30]	0.31 [0.23,0.39]	0.69 [0.62,0.75]	–
	k-medoids	0.38 [0.24,0.51]	**0.69** [0.61,0.75]	**0.70** [0.62,0.76]	–
	greedy variance-reduction	**0.50** [0.34,0.65]	**0.69** [0.61,0.75]	0.69 [0.61,0.76]	–

Table 8: Predictors on MTEB-main: paired difference $R^2_{\text{predictor}} - R^2_{\text{GAUSS}}$ with 95% bootstrap CI (positive = better than GAUSS); the GAUSS rows show its own R^2 [95% CI].

Selection	Predictor	k=1	k=5	k=10	k=20	k=40
random	AvgShift	+0.004 [+0.002,+0.006]	-0.099 [-0.113,-0.087]	-0.136 [-0.154,-0.121]	-0.168 [-0.188,-0.151]	-0.193 [-0.215,-0.173]
	kNN	+0.053 [+0.030,+0.071]	+0.015 [-0.000,+0.029]	+0.018 [-0.001,+0.034]	-0.004 [-0.024,+0.014]	-0.021 [-0.042,-0.000]
	ISVD	+0.003 [-0.009,+0.015]	-0.012 [-0.020,-0.005]	-0.020 [-0.028,-0.013]	-0.029 [-0.036,-0.022]	-0.036 [-0.045,-0.028]
	FA	-0.009 [-0.019,+0.000]	-0.002 [-0.008,+0.004]	-0.007 [-0.013,-0.002]	-0.013 [-0.019,-0.007]	-0.021 [-0.031,-0.010]
	Gauss	0.394 [0.361,0.423]	0.652 [0.618,0.684]	0.720 [0.690,0.749]	0.769 [0.743,0.793]	0.802 [0.781,0.823]
	Hybrid	-0.004 [-0.006,-0.002]	+0.001 [-0.001,+0.002]	+0.002 [+0.001,+0.004]	+0.005 [+0.003,+0.008]	+0.013 [+0.010,+0.018]
greedy variance-reduction	AvgShift	+0.005 [+0.002,+0.008]	-0.128 [-0.146,-0.112]	-0.166 [-0.188,-0.146]	-0.199 [-0.233,-0.172]	-0.201 [-0.227,-0.177]
	kNN	+0.075 [+0.047,+0.100]	-0.007 [-0.030,+0.014]	-0.010 [-0.034,+0.012]	-0.021 [-0.043,+0.001]	-0.020 [-0.043,+0.002]
	ISVD	+0.006 [-0.004,+0.018]	-0.018 [-0.030,-0.008]	-0.023 [-0.032,-0.014]	-0.037 [-0.045,-0.030]	-0.036 [-0.044,-0.027]
	FA	+0.018 [+0.006,+0.029]	+0.000 [-0.010,+0.010]	-0.007 [-0.016,+0.001]	-0.015 [-0.025,-0.005]	-0.013 [-0.027,+0.001]
	Gauss	0.520 [0.479,0.559]	0.709 [0.676,0.741]	0.758 [0.733,0.781]	0.788 [0.767,0.809]	0.800 [0.778,0.820]
	Hybrid	-0.003 [-0.005,-0.001]	+0.002 [+0.001,+0.004]	+0.004 [+0.002,+0.006]	+0.010 [+0.007,+0.012]	+0.023 [+0.018,+0.029]

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Table 9: Where the selection covariance comes from: relative MSE reduction (%, with CI) of greedy selection using the covariance of all rows including test models ("oracle"), or the rank-r factor-model covariance, over our default (pairwise covariance from training rows). Positive = better than default.

Dataset	Selection covariance	k=1	k=3	k=5	k=10	k=20
MTEB-main	oracle covariance	+1.0 [+0.1,+1.9]	+2.6 [+1.1,+4.0]	+3.8 [+1.2,+6.7]	+2.6 [+1.3,+3.9]	+0.8 [-3.6,+3.9]
MTEB-main	FA covariance	+0.2 [-3.4,+3.7]	-1.5 [-4.6,+1.1]	-2.5 [-5.2,-0.2]	-5.3 [-8.3,-2.1]	-9.4 [-13.8,-5.9]
MTEB-dense	oracle covariance	+1.6 [+0.0,+3.2]	+3.0 [+0.8,+5.3]	+2.1 [+0.1,+4.3]	+1.8 [-0.6,+4.1]	+0.4 [-2.0,+2.6]
MTEB-dense	FA covariance	-0.6 [-3.8,+1.9]	-2.2 [-5.1,+0.4]	-1.3 [-4.2,+1.2]	+0.6 [-1.3,+2.6]	-4.0 [-6.1,-1.7]
MTEB-eng	oracle covariance	+3.6 [+1.1,+6.6]	+4.6 [+1.8,+7.4]	+3.8 [+1.2,+6.4]	+1.3 [-1.2,+3.7]	+0.3 [-1.7,+2.4]
MTEB-eng	FA covariance	+1.1 [-6.1,+7.5]	-6.5 [-10.7,-2.6]	+1.2 [-2.5,+4.8]	+2.8 [-0.3,+5.9]	+0.2 [-1.9,+2.3]
HELM-Lite-22	oracle covariance	+2.3 [+0.6,+4.8]	+3.6 [+0.6,+6.8]	+0.9 [-0.4,+2.4]	+0.9 [-0.0,+1.9]	–
HELM-Lite-22	FA covariance	+0.5 [-0.7,+1.9]	-0.9 [-2.8,+0.9]	-0.9 [-2.6,+0.8]	-0.2 [-1.2,+0.6]	–

Table 10: Synthetic factor models, $N = 100$, $k = 8$: relative reduction of the objective F by greedy selection vs. random selection, by rank r, factor strength g (per-benchmark SNR of each factor) and leverage heterogeneity κ (log-sd of loading scales).

r	g	κ=0	κ=0.5	κ=1	κ=1.5
1	0.3	3%	4%	4%	3%
1	1	5%	6%	6%	4%
1	3	8%	8%	7%	6%
2	0.3	8%	8%	7%	4%
2	1	12%	12%	10%	7%
2	3	16%	16%	13%	9%
4	0.3	15%	14%	9%	5%
4	1	24%	22%	16%	9%
4	3	29%	28%	21%	13%
8	0.3	24%	23%	16%	10%
8	1	45%	40%	28%	17%
8	3	57%	54%	38%	25%

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3. **Theory assumptions and proofs**

Table 11: First twelve benchmarks chosen by the greedy rule on the full MTEB-main correlation matrix (illustration, in-sample), and the residual variance per benchmark they leave. The first pick is a multilingual retrieval task, followed by English retrieval, classification and STS: the rule spreads its picks over families rather than over one type.

j	Task	Type / languages	resid. var.
1	BelebeleRetrieval	Retrieval / acm,afr,als...	0.55
2	CQADupstackPhysicsRetrieval	Retrieval / eng	0.43
3	AngryTweetsClassification	Classification / unk	0.38
4	STS14	STS / eng	0.36
5	CyrillicTurkicLangClassification	Classification / bak,chv,kaz...	0.34
6	TempReasonL2Fact	Retrieval / eng	0.32
7	KorSTS	STS / kor	0.31
8	LEMBQMSumRetrieval	Retrieval / eng	0.30
9	PSC	PairClassification / pol	0.29
10	NusaParagraphEmotionClassification	Classification / bbc,bew,bug...	0.27
11	RUParaPhraserSTS	STS / rus	0.27
12	MedrxivClusteringS2S.v2	Clustering / eng	0.26

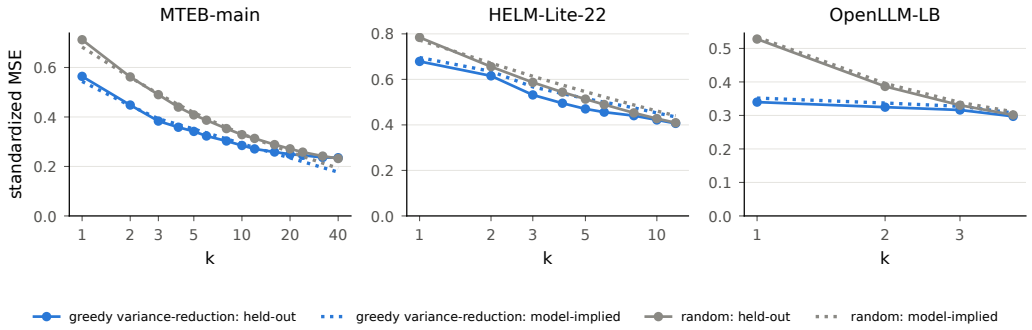


Figure 5: Model-implied expected MSE (dotted) vs. held-out MSE (solid) for greedy and random selection.

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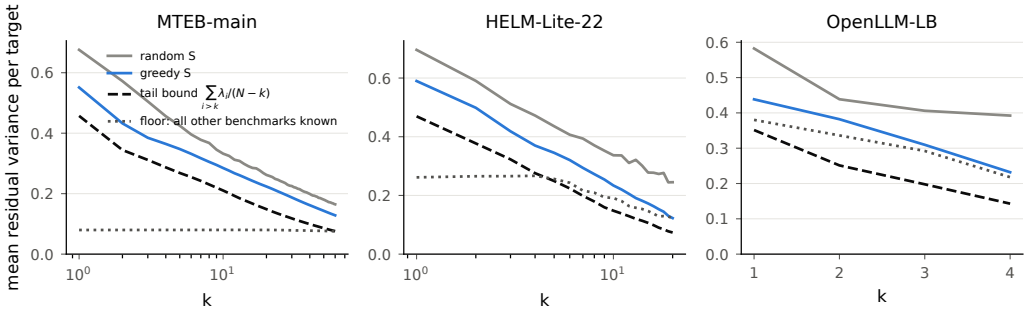


Figure 6: In-sample view on the full matrices (pairwise correlation matrix, light shrinkage): mean residual variance per remaining benchmark for random and greedy sets, the spectral tail bound of Prop. 3, and the floor if all other benchmarks were known. Greedy sits between random and the bound.

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    Justification: The proposed methods use no LLMs. For transparency: the research, code and text were produced by an autonomous AI research agent (Claude) working unattended; all reported numbers were generated by the released code.

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